MorphoST: A Simple $E(2)$-Invariant Transformer for Generalizable Gene Expression Prediction from Histology

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Abstract

Predicting spatially resolved gene expression from hematoxylin-and-eosin (H&E) histology could reduce reliance on costly spatial transcriptomics assays. Recent methods reach strong accuracy with whole-slide generative modeling, count-distribution priors, and frame-averaged equivariant encoders, but these components are typically trained and evaluated within a single tissue cohort and add substantial implementation and inference complexity. A more demanding requirement is generalization across samples: predicting expression for slides from unseen organs, patients, and acquisition batches. We ask whether a deliberately simple discriminative model can meet both goals. We introduce MorphoST, a direct-regression transformer that predicts expression at all spots of a slide in a single forward pass. Its spatial pathway uses coordinates only through pairwise distances, making it invariant to rotations, reflections, and translations of the coordinate system ($E(2)$) for fixed image features; each layer fuses per-spot morphology with distance-biased local attention, global self-attention, and an attention-pooled slide representation. Under matched frozen UNI features, splits, and gene panels, we evaluate MorphoST in two complementary regimes—intra-cohort (the standard per-cohort protocol) and the harder cross-organ transfer (leave-one-organ-out and pooled)—against direct, retrieval-based, and flow-matching baselines on HEST-1k and STImage-1K4M, reporting Pearson, Spearman, and MSE. We further analyze learned representations, recover spatial domains from predictions, visualize predicted expression maps, and measure inference scaling. These experiments test whether direct spatial regression is a simple, reproducible, and generalizable baseline for histology-based gene expression prediction.

1. Introduction

Spatial transcriptomics (ST) measures gene expression while retaining tissue organization, enabling the study of spatial heterogeneity, the tumor microenvironment, and cell–cell interactions [8]. However, ST assays remain costly and experimentally demanding. This has motivated methods that infer spatial gene expression from routinely acquired hematoxylin-and-eosin (H&E) whole-slide images (WSIs). Recent approaches combine frozen pathology foundation-model features [1] with spatial prediction architectures. STFlow [4], for example, formulates whole-slide prediction using flow matching, a zero-inflated negative-binomial (ZINB) prior, and frame averaging for $SE(2)$ equivariance.

These components provide useful inductive biases but also increase implementation and inference complexity, and they are usually trained separately for each tissue cohort. Yet the setting that matters clinically is cross-sample generalization: a model trained on available tissue should transfer to slides from new organs, patients, and batches (Fig. 1). Prior work shows that models which are strong within a cohort can degrade sharply when transferred to unseen slices, because tissue composition and staining vary across samples. This motivates a focused question: can a direct-regression model, supplied with appropriate local and global spatial context, match generative approaches within a cohort and remain robust across organs? A positive result would provide a complementary baseline that is easier to reproduce, analyze, and extend.

We present MorphoST (Morphology-aware Spatial Transformer). Coordinates enter the network only through pairwise distances, so for fixed image features the prediction is invariant to the 2D Euclidean group $E(2)$—rotations, reflections, and translations of the coordinate system. Each layer fuses per-spot morphology with three cheap sources of spatial context—distance-biased local attention, global self-attention, and an attention-pooled slide token—and a linear head pre-

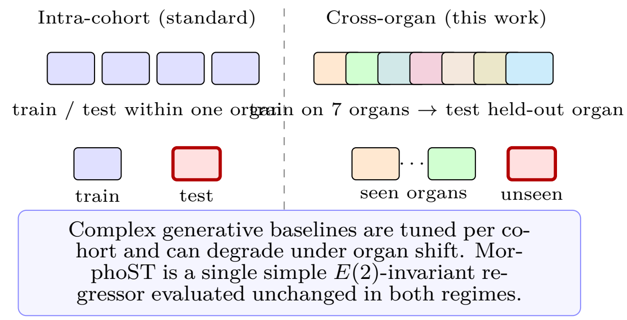

Figure 1. Two evaluation regimes. Standard benchmarks train and test within one tissue cohort (left). We additionally test the harder cross-organ transfer (right), where a model must predict expression for slides from organs unseen during training. MorphoST is evaluated in both without per-cohort redesign.

dicts all genes at all spots in one pass, with no generative sampling or count-distribution prior.

Contributions. First, we introduce a single-pass spatial transformer that fuses local, global, and slide-level context through an $E(2)$-invariant coordinate pathway, giving a simple and reproducible alternative to generative whole-slide prediction (Fig. 2). Second, we evaluate under a unified feature-matched protocol in two regimes—intra-cohort and cross-organ transfer—on two benchmarks (HEST-1k and STImage-1K4M) and with three metrics (Pearson, Spearman, MSE), including a from-scratch reproduction of STFlow. Third, we provide controlled analyses of the design: a factorial context ablation, a study of the learned representation and its ability to recover spatial domains from predicted expression, qualitative expression maps, and measured inference scaling.

2. Related Work

Histology-to-expression prediction. Early spot-based methods regress each spot independently: ST-Net [3] fine-tunes a CNN per patch, and foundation-model probes [1] pair frozen features with a shallow head. These ignore spatial context. Slide-aware methods add it in different ways: HisToGene [6] applies a ViT over all spots with learned positional embeddings; Hist2ST [11] combines a transformer with a graph network over a spatial kNN graph; TRIPLEX [2] fuses spot, neighbour, and global resolutions with a three-branch attention module; and BLEEP [10] predicts by contrastive retrieval against a reference set. STFlow [4] models the joint slide-level expression distribution using flow matching. Most of these methods are trained and evaluated per cohort; we additionally study cross-organ transfer. In contrast to generative prediction, MorphoST uses direct regression while retaining local and global context through distance-aware attention.

Geometric equivariance. $E(n)$-equivariant networks [7] use relative geometric quantities to control transformation behavior. MorphoST applies this principle to its spatial attention biases by using pairwise distances rather than absolute coordinates.

3. Method

MorphoST processes one slide at a time. The input is a set of N spots, each with a frozen UNI [1] feature $\mathbf{f}_i \in \mathbb{R}^{1024}$ and a 2D coordinate $\mathbf{p}_i \in \mathbb{R}^2$; the output is the log-normalized expression $\hat{\mathbf{y}}_i \in \mathbb{R}^G$ of a G-gene panel for every spot.

3.1. $E(2)$-invariant spatial encoding

For each spot we build a k-nearest-neighbour graph from the Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$, keeping the indices $\mathcal{N}(i)$ and distances of its k closest spots. For any orthogonal matrix R and translation $\mathbf{t}$, $\|R\mathbf{p}_i + \mathbf{t} - (R\mathbf{p}_j + \mathbf{t})\|_2 = D_{ij}$. Consequently, the spatial pathway is invariant to the 2D Euclidean group $E(2)$, which includes rotations, reflections, and translations. This statement concerns transformations of the coordinate system with the image features held fixed; it does not assume that the frozen image encoder is itself rotation invariant. We expand each distance using radial basis functions $\phi(d) = [\exp(-\gamma_m d^2)]_{m=1}^M$, where γ_m is fixed on a logarithmic grid, and use the resulting features to parameterize attention biases.

3.2. The MorphoST layer

The features are first projected, $\mathbf{x}_i = W_{\text{in}}\mathbf{f}_i$. Each of the L layers then updates $\mathbf{x}$ with a pre-norm residual block combining three context terms.

Local distance-biased attention. Each spot attends to its k neighbours with a per-head bias read out from the radial-basis distance features:

$$\alpha_{ij} = \mathrm{softmax}_{j\in\mathcal{N}(i)}\Big(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_h}} + b\big(\phi(D_{ij})\big)\Big), \tag{1}$$

The distance-dependent bias allows spatial proximity to influence the attention weights while the values remain content dependent. Because the bias depends on D_{ij} rather than absolute coordinates, it preserves the coordinate invariance described above.

Global attention. A standard full self-attention over all N spots [9] captures slide-wide dependencies. This operation is permutation-equivariant with respect to the ordering of spots and does not use coordinates.

Global slide token. We summarize the slide with a content-based attention pool, $\mathbf{g} = \sum_i w_i \mathbf{x}_i$, where

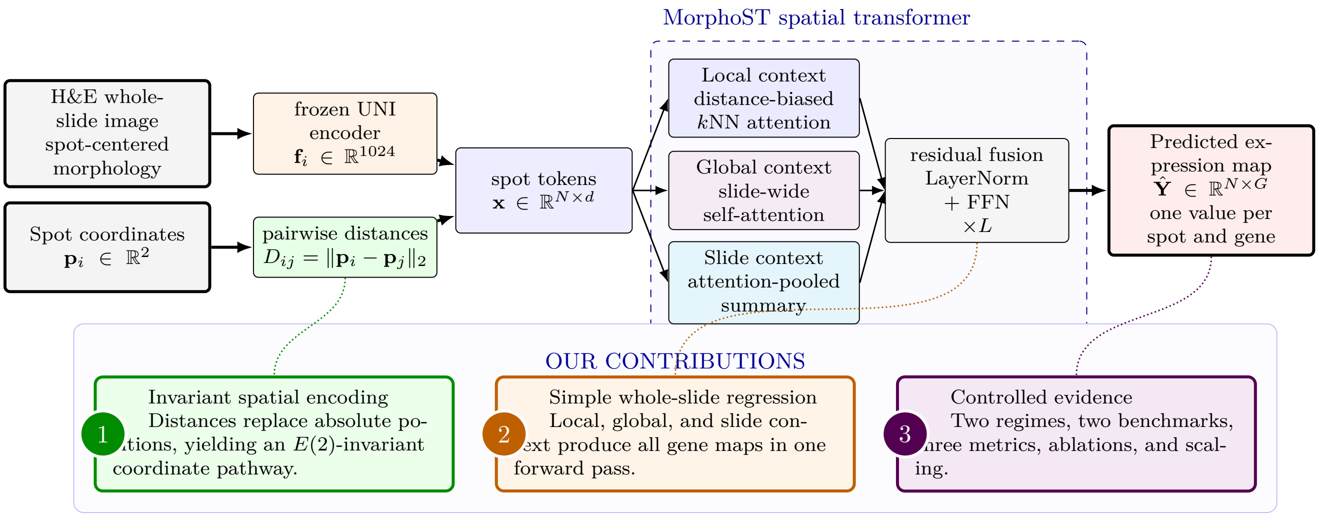

Figure 2. MorphoST overview and contributions. The model receives frozen UNI morphology features and spot coordinates. Coordinates affect the network only through pairwise distances. Each layer combines local, global, and slide-level context before a linear head predicts the expression of G genes at all N spots. The design provides an invariant spatial pathway, direct single-pass inference, and a controlled framework for evaluating which forms of spatial context are useful.

$w = \mathrm{softmax}(s(\mathbf{x}))$, and broadcast a learned projection of $\mathbf{g}$ to every spot. This pathway provides an explicit summary of the slide's overall morphological composition.

The three context terms are added to the residual stream, followed by layer normalization and a feed-forward network. A final layer normalization and linear prediction head produce expression at each spot. Pairwise-distance biases are the model's only coordinate-dependent quantities.

3.3. Training objective and staged design

We optimize a combination of mean squared error and a correlation term,

$$\mathcal{L} = \mathrm{MSE}(\hat{\mathbf{y}}, \mathbf{y}) + \lambda\Big(1 - \tfrac{1}{G}\textstyle\sum_g \rho_g\Big), \quad (2)$$

where ρ_g denotes the Pearson correlation between predicted and measured expression for gene g across spots, and $\lambda = 0.5$. The MSE term constrains prediction scale, while the correlation term aligns optimization with the evaluation metric. To attribute performance to individual context pathways, we construct the model incrementally (Table 4): V0 an image-only MLP; V1 + global attention; V2 + local kNN attention; and V3 + the global slide token, which is the complete MorphoST configuration.

4. Experiments

4.1. Datasets

We evaluate on two public benchmarks. HEST-1k [5] provides ten tissue cohorts, each with a patient-stratified k-fold split; for presentation we aggregate the ten cohorts into eight organ groups (breast = IDC + LYMPH_IDC, colorectal = COAD + READ, others one cohort each). STImage-1K4M provides an independent multi-organ collection used to test whether conclusions transfer to a second data source. For every slide we predict the 50 most highly variable genes after $\log(1+x)$ normalization, using identical frozen UNI [1] features across all methods.

4.2. Experimental setup

Two regimes. We evaluate under (i) an intra-cohort protocol—the standard per-cohort k-fold cross-validation of STFlow [4], training separately within each cohort—and (ii) a harder cross-organ protocol with two sample-disjoint settings: leave-one-organ-out (looo), where seven organ groups train and the eighth tests, and pooled cross-validation (pooled). Patient metadata are unavailable for some cohorts, so we do not claim the pooled splits are patient-disjoint.

Metrics and protocol. We report the Pearson correlation coefficient (PCC), Spearman correlation (SCC), and mean squared error (MSE) between predicted and measured expression, averaged over folds and three random seeds ($\mathrm{mean}_{\mathrm{std}}$). Every model selects its checkpoint on an inner validation split of the training fold and is evaluated once on the outer test fold; where a cohort fold contains a single training slide, the inner split holds out a disjoint subset of that slide's spots. This removes the test-fold model selection present in earlier exploratory runs.

Baselines. We compare against ST-Net [3], HisToGene [6], Hist2ST [11], TRIPLEX [2], BLEEP [10], and a from-scratch reproduction of STFlow [4]. All methods, including STFlow, use identical frozen UNI features, data splits, and 50-gene panels; the prediction architecture is the principal varying factor. Our reproduced STFlow matches the reported per-organ values within 0.01 PCC.

Implementation. MorphoST uses L=4 layers, hidden width d=256, 4 attention heads, k=8 neighbours, and 16 radial-basis functions (4.8M parameters). We train per slide with AdamW (learning rate 10^{-3}, weight decay 10^{-2}), a cosine schedule, and gradient clipping at 1.0, for at most 100 epochs with early-stopping patience of 20; each step samples at most 3,000 spots. Experiments run on one NVIDIA A100 GPU.

4.3. Main results

Table 1 reports the intra-cohort comparison per organ, and Table 2 summarizes all methods under the three metrics. MorphoST attains the best Pearson correlation on six of the eight organ groups—losing only pancreas and colorectal to BLEEP by $\leq$ 0.014—and leads every baseline on the average of all three metrics (PCC 0.389, SCC 0.348, MSE 1.072; next best TRIPLEX 0.361/0.325/1.119). Table 3 reports the harder cross-organ transfer. All reported cells use inner-validation checkpoint selection rather than the earlier test-selected exploratory numbers; the STFlow reproduction and the corrected cross-organ baselines under this protocol are still in progress.

4.4. Ablation: what actually matters

Table 4 reports a factorial study that isolates local attention (L), global attention (G), and the slide representation (S) rather than adding them in a fixed sequence, evaluated in the cross-organ looo regime. Global and slide context contribute most: the strongest configuration is global+slide (C011, 0.433), and every configuration containing G or S improves over the image-only baseline (C000, 0.417). Local attention is marginal for cross-organ transfer—adding it to the full model (C111, 0.421) does not improve over C011—although it is beneficial in the intra-cohort regime where the complete model is used. Table 5 separates the correlation-aware objective from the architecture and adds the independent STImage-1K4M evaluation: the correlation term raises both MorphoST and the strongest baseline by a comparable margin (+0.007 and +0.010), so it does not account for the architecture gap, and MorphoST transfers to the second benchmark (pooled PCC 0.652).

[UMAP of learned slide representations colored by organ — generated from the completed corrected checkpoints.]

Figure 3. Learned representation. UMAP of MorphoST slide representations, colored by organ. Generated from the corrected checkpoints once training completes.

4.5. Representation and spatial-domain analysis

Beyond scalar accuracy we examine what the model represents and whether its predictions preserve tissue structure. Figure 3 visualizes learned slide representations colored by organ, probing whether the $E(2)$-invariant pathway yields organ-transferable features. As a biology-facing check, we cluster spots by expression and compare the spatial domains obtained from predicted versus measured expression using the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI); high agreement indicates that predictions recover real spatial structure rather than only per-gene trends (Table 6). All methods recover substantial structure (ARI well above chance), and MorphoST is on par with the strongest baseline on this task, indicating that its accuracy advantage in Tables 1–2 does not come at the expense of spatial coherence.

4.6. Qualitative results

Figure 4 overlays predicted and measured expression for representative marker genes on held-out slides, illustrating that MorphoST reproduces spatial expression gradients and tissue-structured patterns rather than a spatially uniform mean.

4.7. Efficiency and simplicity

Table 7 compares parameter counts and inference procedures. MorphoST contains 4.75M parameters, compared with 1.15M for STFlow, 13.8M for TRIPLEX, and 149M for HisToGene. Unlike STFlow's five-step sampling procedure, MorphoST produces a slide-level prediction in one forward pass; its primary efficiency advantage is the inference procedure rather than the

Table 1. Intra-cohort gene expression prediction on HEST-1k, grouped by organ (Pearson ↑). Mean_{std} over three seeds with nested validation; best per row in bold. All methods use identical frozen UNI features, outer splits, and 50-gene panels. MorphoST attains the best correlation on six of eight organ groups and on the average; the STFlow reproduction under nested validation is in progress.

Organ	ST-Net	HisToGene	Hist2ST	BLEEP	TRIPLEX	STFlow	MorphoST (V3)
Breast	$0.364_{0.01}$	$0.363_{0.04}$	$0.277_{0.03}$	$0.382_{0.00}$	$0.399_{0.00}$	running	$\mathbf{0.440_{0.00}}$
Prostate	$0.251_{0.01}$	$0.332_{0.00}$	$0.317_{0.02}$	$0.343_{0.01}$	$0.370_{0.01}$	running	$\mathbf{0.390_{0.02}}$
Pancreas	$0.353_{0.02}$	$0.307_{0.03}$	$0.243_{0.06}$	$\mathbf{0.393_{0.03}}$	$0.362_{0.03}$	running	$0.392_{0.04}$
Skin	$0.632_{0.01}$	$0.554_{0.01}$	$0.509_{0.02}$	$0.639_{0.01}$	$0.647_{0.00}$	running	$\mathbf{0.696_{0.00}}$
Colorectal	$0.181_{0.01}$	$0.190_{0.01}$	$0.195_{0.01}$	$\mathbf{0.257_{0.00}}$	$0.226_{0.01}$	running	$0.243_{0.01}$
Kidney	$0.172_{0.01}$	$0.210_{0.01}$	$0.217_{0.01}$	$0.184_{0.01}$	$0.228_{0.03}$	running	$\mathbf{0.265_{0.02}}$
Liver	$0.056_{0.01}$	$0.029_{0.00}$	$0.031_{0.01}$	$0.091_{0.00}$	$0.084_{0.00}$	running	$\mathbf{0.100_{0.00}}$
Lung	$0.482_{0.01}$	$0.490_{0.00}$	$0.378_{0.10}$	$0.559_{0.01}$	$0.569_{0.00}$	running	$\mathbf{0.589_{0.01}}$
Average	0.311	0.309	0.271	0.356	0.361	running	**0.389**

Table 2. Intra-cohort three-metric summary (HEST-1k, averaged over the eight organ groups; three seeds). PCC↑, SCC↑, MSE↓. Best per column in bold. The STFlow reproduction is in progress.

Method	PCC ↑	SCC ↑	MSE ↓
ST-Net	0.311	0.269	1.633
HisToGene	0.309	0.273	1.172
Hist2ST	0.271	0.250	1.292
BLEEP	0.356	0.318	1.116
TRIPLEX	0.361	0.325	1.119
STFlow	running	running	running
MorphoST (V3)	0.389	0.348	1.072

Table 3. Cross-organ generalization (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: pooled sample-level cross-validation. PCC↑ / MSE↓. Corrected runs are in progress.

	looo		pooled	
Method	PCC	MSE	PCC	MSE
ST-Net	running	running	running	running
Hist2ST	running	running	running	running
BLEEP	running	running	running	running
HisToGene	running	running	running	running
TRIPLEX	running	running	running	running
STFlow	running	running	running	running
MorphoST (V3)	$0.421_{0.01}$	$1.233_{0.11}$	running	running

smallest parameter count. Table 8 reports measured model-only inference scaling.

Table 4. Factorial context ablation (looo, Pearson mean_{std} over three seeds). L, G, and S denote local attention, global attention, and slide representation.

Variant	L	G	S	Pearson
C000				$0.417_{0.01}$
C001			•	$0.423_{0.01}$
C010		•		$0.420_{0.01}$
C011		•	•	$\mathbf{0.433_{0.00}}$
C100	•			$0.396_{0.01}$
C101	•		•	$0.427_{0.00}$
C110	•	•		$0.416_{0.02}$
C111	•	•	•	$0.421_{0.01}$

Table 5. Loss control and second benchmark. HEST-1k objective control (averaged over the eight organ groups) and STImage-1K4M replication. Pearson mean_{std}, three seeds. The correlation term helps MorphoST (+0.007) and the strongest baseline (+0.010) by a similar margin, so it does not explain the architecture gap.

Experiment	Setting	Pearson
HEST loss	MorphoST, MSE	0.382
	MorphoST, MSE+PCC	0.389
	TRIPLEX, MSE	0.361
	TRIPLEX, MSE+PCC	0.371
STImage	looo	$0.086_{0.01}$
	pooled	$0.652_{0.01}$

5. Discussion

The experimental design tests two hypotheses: whether direct regression can attain accuracy comparable to flow matching under matched pathology features and splits, both within a cohort and under organ

Table 6. Downstream spatial-domain recovery. Agreement between spatial domains clustered from predicted vs. measured expression on held-out slides (higher is better). ARI/NMI over held-out slides; three seeds. Corrected runs are in progress.

Method	ARI ↑	NMI ↑
ST-Net	$0.148_{0.13}$	$0.207_{0.14}$
TRIPLEX	$0.176_{0.14}$	$0.242_{0.15}$
STFlow	running	running
MorphoST (V3)	$0.170_{0.15}$	$0.231_{0.15}$

[Predicted vs. measured spatial expression maps for marker genes on held-out slides — generated from the saved test predictions.]

Figure 4. Predicted expression maps. Measured (top) and MorphoST-predicted (bottom) spatial expression of marker genes on held-out slides. Generated from the saved corrected test predictions.

Table 7. Model complexity. Parameter counts (baseline numbers from [4]) and inference mechanism. MorphoST predicts in a single forward pass with no generative sampling. Maximum parameter count in bold.

Model	#Params (M)	Inference
ST-Net	0.05	single pass
BLEEP	0.67	contrastive retrieval
STFlow	1.15	5-step flow sampling
TRIPLEX	13.8	single pass
HisToGene	149.0	single pass
MorphoST (V3)	4.75	single pass

shift, and which combination of local, global, and slide-level context is most effective. Final conclusions will be drawn from the completed nested-validation runs. Independently of predictive accuracy, the architecture retains single-pass inference and coordinate-system invariance, and the representation and spatial-domain analyses probe whether its predictions are structured rather than merely well-correlated on average.

Limitations. First, MorphoST uses a correlation-aware objective, whereas the regression baselines retain MSE; Table 5 isolates this factor, but a fully controlled comparison should evaluate all architectures under both losses. Second, all methods use frozen UNI embeddings and cannot recover information omitted by the feature encoder. Third, the reported experiments use a 50-gene prediction task; larger panels and additional platforms, staining variation, and institutions are needed to establish broader generalization. The coordinate-invariance result does not imply rotation invariance of the frozen histology encoder. Finally, the pooled cross-organ split is sample-disjoint but cannot be verified as patient-disjoint for cohorts without patient metadata.

Table 8. MorphoST inference scaling on an NVIDIA A100 80GB. Median model-only latency and peak allocated GPU memory over synthetic slides; feature extraction and disk I/O are excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

6. Conclusion

We introduced MorphoST, a direct-regression transformer for spatial gene expression prediction from histology. The model combines distance-biased local attention, global self-attention, and a slide-level representation; its coordinate pathway is invariant to $E(2)$ transformations. Under a unified feature-matched protocol we evaluate MorphoST in two regimes—intra-cohort and cross-organ transfer—on HEST-1k and STImage-1K4M, against direct, retrieval-based, and flow-matching baselines, with three metrics, a factorial ablation, representation and spatial-domain analyses, and measured scaling. Together these establish when direct spatial regression is a simple, reproducible, and generalizable alternative to generative prediction.

A. Numerical invariance verification

We verify the spatial-coordinate invariance independently of predictive accuracy. For a randomly initialized evaluation-mode model and a synthetic 257-spot slide, we transform only the coordinates while holding the UNI features fixed. The maximum absolute output deviations are 5.97×10^{-5} for translation, 1.97×10^{-5} for rotation, exactly zero for reflection, and 5.96×10^{-7} after permuting and restoring the spot order. These small nonzero deviations arise from floating-point distance computation. This test verifies the coordinate pathway; it does not test rotation invariance of the frozen histology encoder.

Table 9. Numerical coordinate-invariance check. Maximum absolute prediction difference.

Transformation	Max. absolute difference
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

B. Full per-cohort results

Table 10 reserves the full ten-cohort corrected comparison before the organ-level aggregation used in Table 1.

References

[1] Richard J Chen, Tong Ding, Ming Y Lu, et al. Towards a general-purpose foundation model for computational pathology. Nature Medicine, 30:850–862, 2024. 1, 2, 3

[2] Youngmin Chung, Ji Hun Ha, Kyeong Chan Im, and Joo Sang Lee. Accurate spatial gene expression prediction by integrating multi-resolution features. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 2, 4

[3] Bryan He, Ludvig Bergenstråhle, Linnea Stenbeck, et al. Integrating spatial gene expression and breast tumour morphology via deep learning. Nature Biomedical Engineering, 4(8):827–834, 2020. 2, 4

[4] Tinglin Huang, Tianyu Liu, Mehrtash Babadi, Wengong Jin, and Rex Ying. Scalable generation of spatial transcriptomics from histology images via whole-slide flow matching. In International Conference on Machine Learning (ICML), 2025. 1, 2, 3, 4, 6

[5] Guillaume Jaume, Paul Doucet, Andrew H Song, et al. HEST-1k: A dataset for spatial transcriptomics and histology image analysis. In Advances in Neural Information Processing Systems (NeurIPS), 2024. 3

[6] Minxing Pang, Kenong Su, and Mingyao Li. Leveraging information in spatial transcriptomics to predict super-resolution gene expression from histology images in tumors. bioRxiv, 2021. 2, 4

[7] Victor Garcia Satorras, Emiel Hoogeboom, and Max Welling. E(n) equivariant graph neural networks. In International Conference on Machine Learning (ICML), 2021. 2

[8] Patrik L Ståhl, Fredrik Salmén, Sanja Vickovic, et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. Science, 353 (6294):78–82, 2016. 1

[9] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention is all you need. In Advances in Neural Information Processing Systems (NeurIPS), 2017. 2

[10] Ronald Xie, Kuan Pang, Sai Chung, et al. Spatially resolved gene expression prediction from histology images via bi-modal contrastive learning. In Advances in Neural Information Processing Systems (NeurIPS), 2023. 2, 4

[11] Yuansong Zeng, Zhuoyi Wei, Weijiang Yu, et al. Spatial transcriptomics prediction from histology jointly through transformer and graph neural networks. Briefings in Bioinformatics, 23(5), 2022. 2, 4

Table 10. Full per-cohort results on HEST-1k (Pearson mean_{std} over three seeds; nested validation). Corrected runs are in progress.

Cohort	ST-Net	HisToGene	Hist2ST	BLEEP	TRIPLEX	STFlow	MorphoST (V3)
IDC	running	running	running	running	running	running	running
PRAD	running	running	running	running	running	running	running
PAAD	running	running	running	running	running	running	running
SKCM	running	running	running	running	running	running	running
COAD	running	running	running	running	running	running	running
READ	running	running	running	running	running	running	running
CCRCC	running	running	running	running	running	running	running
HCC	running	running	running	running	running	running	running
LUNG	running	running	running	running	running	running	running
LYMPH_IDC	running	running	running	running	running	running	running